

Phase 1 Energy: Discretization of the Γ -Convergence Functional

Dirichlet term, double well, Modica–Mortola limit, and the crispness penalty

July 2026

Abstract

This document derives the discretized Phase 1 relaxation energy minimized by `ProjectedGradientOptimizer` (`src/optimization/pgd_optimizer.py`) and its gradient, exactly as implemented after the double-well correction of 2026-07-08. The continuous functional is the Modica–Mortola-type Γ -convergence energy of Bogosel & Oudet: a Dirichlet interface term plus a double well, whose $\varepsilon \rightarrow 0$ limit is the partition perimeter, augmented by a soft *crispness* penalty. Three quantities are derived from the P_1 finite-element discretization: the Dirichlet term $\varepsilon u^\top K u$, the double-well term $\varepsilon^{-1} q^\top M q$ with $q = u(1 - u)$, and the variance penalty $\lambda(1 - \text{Var}/T)$; all three gradients are *analytical* and match the code to machine precision. A remark records the one-token discretization bug this document’s corrected form replaces (the well was previously coded as $q = u^2(1 - u)^2$; see `docs/reference/phase1_energy_discretization_bug.md`).

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1 Overview

Phase 1 of the method (relaxation) replaces the geometric problem “partition the surface S into N equal-area regions of minimal total interface length” by a smooth minimization over *density fields*. Each region k is represented by a function $u_k : S \rightarrow [0, 1]$, where $u_k(x)$ is “how strongly cell k claims the point x .” The optimizer minimizes a relaxed energy E over the stacked field $u = (u_1, \dots, u_N)$ subject to a partition-of-unity constraint $\sum_k u_k \equiv 1$ and an equal-area constraint $\int_S u_k dA = |S|/N$ for each k (both enforced by projection; `src/optimization/projection.py`). As the interface width $\varepsilon \rightarrow 0$ the relaxed energy Γ -converges to the partition perimeter, so its minimizers approximate minimal-perimeter partitions [1, 2].

This document derives, from the P_1 finite-element discretization, the three additive pieces of E and their gradients *as implemented*. It is the formal companion to the reference note `docs/reference/phase1_energy_discretization_bug.md`, which records that the double-well term was for a long time mis-discretized (a typo copied from the source paper) and was corrected on 2026-07-08; the form derived here is the corrected one. Per the `docs/math` scope policy, every formula below is a faithful transcription of the live code.

2 Notation and setup

Remark 2.1 (The symbol λ). In this document λ denotes the *crispness-penalty weight* (`lambda_penalty`, Section 6). It is unrelated to the Phase 2 variable-point parameter λ of the same name used elsewhere in `docs/math`; Phase 1 has no variable points.

Mesh and P_1 finite elements. S is triangulated with vertices $\mathbf{V} = \{x_1, \dots, x_{|V|}\}$ and triangles **F**. Let $\{\varphi_j\}_{j=1}^{|V|}$ be the piecewise-linear (P_1) nodal “hat” basis, $\varphi_j(x_i) = \delta_{ij}$. A nodal vector $w \in \mathbb{R}^{|V|}$ defines the P_1 function $w_h = \sum_j w_j \varphi_j$, i.e. the linear interpolant of the values w_j . The two standard P_1 assembly matrices are the *mass* and *stiffness* matrices

$$M_{jk} = \int_S \varphi_j \varphi_k dA, \quad K_{jk} = \int_S \nabla_\tau \varphi_j \cdot \nabla_\tau \varphi_k dA, \quad (1)$$

where ∇_τ is the tangential (surface) gradient. Both are built by `TriMesh.compute_matrices` (`src/mesh/tri_mesh.py`); M and K are symmetric and sparse. The *lumped mass* vector is the row sum of M ,

$$v_j = \sum_k M_{jk} = \int_S \varphi_j dA > 0, \quad W = \sum_j v_j = \int_S 1 dA = |S|, \quad (2)$$

so v_j is the area weight of vertex j and W is the total surface area (`self.v`, `self.W` in the optimizer). The equal-area target per cell is $\bar{A} = W/N$ and the target density mean is $\mu_t = 1/N$.

Interface width. The width is tied to the mesh, $\varepsilon = \sqrt{\text{mean triangle area}} \approx h$ (`src/pipeline/relaxation.py`, `_setup_level`), and is held fixed within a refinement level.

The two exact P_1 identities. For any nodal vector w ,

$$\int_S |\nabla_\tau w_h|^2 dA = w^\top K w, \quad (3)$$

$$\int_S w_h^2 dA = w^\top M w, \quad (4)$$

both immediate from (1) and bilinearity, since $w_h = \sum_j w_j \varphi_j$. These are the only two discretization identities the energy uses; everything else is exact algebra.

The continuous functional. Per cell, the relaxed energy is the Modica–Mortola-type functional

$$F_\varepsilon(u_k) = \varepsilon \int_S |\nabla_\tau u_k|^2 dA + \frac{1}{\varepsilon} \int_S W(u_k) dA, \quad W(s) = s^2(1-s)^2, \quad (5)$$

with the standard quartic double well W (minima at $s = 0$ and $s = 1$). The total Phase 1 energy sums this over the N cells and adds the penalty of Section 6:

$$E(u) = \sum_{k=1}^N F_\varepsilon(u_k) + \lambda \sum_{k=1}^N P(u_k). \quad (6)$$

3 The Dirichlet (interface-width) term

Discretizing u_k by its P_1 interpolant and applying (3) gives the interface-width term exactly:

$$\varepsilon \int_S |\nabla_\tau u_k|^2 dA = \varepsilon u_k^\top K u_k = E_k^{\text{dir}}. \quad (7)$$

Because K is symmetric, the gradient in the nodal unknowns u_k is

$$\nabla_{u_k} E_k^{\text{dir}} = 2\varepsilon K u_k. \quad (8)$$

Code: Dirichlet term and gradient

```
src/optimization/pgd_optimizer.py: compute_energy / compute_gradient (per-cell loop)
grad_term = epsilon * (u.T @ (K @ u))
grad_grad = 2 * epsilon * (K @ u)
```

4 The double-well term (the corrected discretization)

The interface (binarization) term is $\varepsilon^{-1} \int_S W(u_k) dA = \varepsilon^{-1} \int_S u_k^2(1-u_k)^2 dA$. Write the integrand as a perfect square,

$$W(u_k) = u_k^2(1-u_k)^2 = [u_k(1-u_k)]^2 = q_k^2, \quad q_k := u_k(1-u_k). \quad (9)$$

Discretizing the *new* field q_k nodally — i.e. forming the vector with entries $q_{k,j} = u_{k,j}(1-u_{k,j})$ and taking its P_1 interpolant — and applying the mass-matrix identity (4) gives

$$\frac{1}{\varepsilon} \int_S W(u_k) dA \approx \frac{1}{\varepsilon} \int_S (q_k)_h^2 dA = \frac{1}{\varepsilon} q_k^\top M q_k = E_k^{\text{well}}. \quad (10)$$

Remark 4.1 (Nodal interpolation of the well). The equality (4) is exact, but (10) carries one approximation: the integrand $q_k = u_k(1-u_k)$ is evaluated at the nodes and then interpolated linearly, so $(q_k)_h$ is the P_1 interpolant of the nodal products, not the exact quadratic $u_h(1-u_h)$. This is the usual nodal-quadrature treatment of a polynomial nonlinearity; it is consistent (error $O(h^2)$) and is what the code computes. Equivalently, the *discrete* double-well energy is defined as $\varepsilon^{-1} \|(q_k)_h\|_{L^2}^2$.

Gradient. Since $q_k = u_k \odot (1-u_k)$ acts componentwise, its Jacobian with respect to u_k is diagonal:

$$\frac{\partial q_{k,j}}{\partial u_{k,j}} = 1 - 2u_{k,j} \quad \implies \quad \frac{\partial q_k}{\partial u_k} = \text{diag}(1 - 2u_k). \quad (11)$$

With M symmetric, the chain rule applied to $q_k^\top M q_k$ gives

$$\nabla_{u_k} E_k^{\text{well}} = \frac{1}{\varepsilon} \text{diag}(1 - 2u_k) (2Mq_k) = \frac{2}{\varepsilon} (Mq_k) \odot (1 - 2u_k), \quad (12)$$

where $\odot$ is the Hadamard (elementwise) product.

Code: double-well term and gradient (corrected)

```
src/optimization/pgd_optimizer.py: compute_energy / compute_gradient (per-cell loop)
interface_vec = u * (1 - u)                                     (= q_k; not u**2*(1-u)**2)
interface_term = (1/epsilon) * (interface_vec.T @ (M @ interface_vec))
grad_interface = (2/epsilon) * (M @ interface_vec) * (1 - 2*u)
```

Remark 4.2 (The discretization bug this form corrects). Earlier code set `interface_vec = u**2 * (1-u)**2`, i.e. $q_k \mapsto q_k^2$. That transcribes a *typo* in the source paper [1] (§3), whose printed definition of the discretized field has an extra squaring, while its prose target integral $\int u^2(1-u)^2$ and its printed gradient $2Mv \odot (1-2u)$ both require $v = u(1-u)$. The buggy energy computed $\varepsilon^{-1} \int (u^2(1-u)^2)_h^2 = \varepsilon^{-1} \int u^4(1-u)^4$ — an eighth-degree well, $\sim 25\times$ too small in magnitude — and its gradient (12) was then no longer the gradient of the coded energy (finite-difference relative error ≈ 1.2). With the correct $q_k = u_k(1-u_k)$, energy (10) and gradient (12) are mutually consistent to machine precision (FD relative error $\approx 10^{-8}$). See `docs/reference/phase1.energy_discretization_bug.md` for the full audit and the $\sim 25\times$ scale shift, which required re-tuning λ .

5 The Modica–Mortola limit: why this is a perimeter

The pairing in (5) is the classical Modica–Mortola functional. For a well $W \geq 0$ vanishing exactly at 0 and 1, the family F_ε Γ -converges as $\varepsilon \rightarrow 0$ (in L^1) to

$$F_0(u) = c_W \cdot \text{Per}(\{u = 1\}), \quad c_W = 2 \int_0^1 \sqrt{W(s)} \, ds, \quad (13)$$

finite only on fields $u \in \{0, 1\}$ of finite perimeter [2]. For the quartic well $W(s) = s^2(1-s)^2$ we have $\sqrt{W(s)} = s(1-s)$ on $[0, 1]$, so

$$c_W = 2 \int_0^1 s(1-s) \, ds = 2 \left[\frac{1}{2} - \frac{1}{3} \right] = \frac{1}{3}. \quad (14)$$

Summing over the N cells as in (6), each interface between two adjacent regions lies in the boundary of *two* phases and is therefore counted twice, so the limit of $\sum_k F_\varepsilon(u_k)$ is $2c_W = \frac{2}{3}$ times the total length of the partition’s interface network. The constant is immaterial to the minimizer: minimizing the relaxed energy at small ε drives the partition toward minimal total interface length — the perimeter problem the method targets. This is the sense in which the coefficient balance $\varepsilon u^\top K u$ versus $\varepsilon^{-1} q^\top M q$ matters: the double-well correction of Section 4 restores W to the quartic (5) whose limit constant is (14), rather than an eighth-degree well with a different (and much flatter) profile.

6 The crispness penalty

The double well rewards *pointwise* binarization but does not by itself force a cell to concentrate its (area-constrained) mass into a compact, sharp region. The optimizer adds a soft penalty P that rewards each cell’s density having the *variance of a sharp 0/1 indicator* of the target size. All statistics are *area-weighted* by the lumped mass v of (2).

Weighted mean and variance. For a cell density u (dropping the cell index),

$$\mu(u) = \frac{v^\top u}{W}, \quad \text{Var}(u) = \frac{1}{W} \sum_j v_j (u_j - \mu)^2 = \frac{((u - \mu \mathbf{1}) \odot v)^\top (u - \mu \mathbf{1})}{W}. \quad (15)$$

On the constraint set $v^\top u = \bar{A}$ the mean is pinned at $\mu = \bar{A}/W = 1/N = \mu_t$.

Target variance. A sharp indicator equal to 1 on an area fraction μ_t and 0 elsewhere has area-weighted variance $\mu_t(1 - \mu_t)$. The penalty compares $\text{Var}(u)$ to a target

$$T = \begin{cases} \mu_t(1 - \mu_t), & \text{fixed target (penalty_target_mode = 'fixed')}, \\ \mu(1 - \mu), & \text{adaptive target ('adaptive')}, \end{cases} \quad T_{\text{eff}} = T + \varepsilon_P, \quad (16)$$

with a small floor $\varepsilon_P = \text{penalty_eps}$ guarding the division. The penalty per cell is

$$P(u) = \lambda \left(1 - \frac{\text{Var}(u)}{T_{\text{eff}}} \right). \quad (17)$$

It is minimized (drops toward $\lambda \cdot 0$) as $\text{Var}(u) \rightarrow T$, i.e. as the cell becomes as “crisp” as a sharp indicator of the right size; a diffuse, low-variance density is penalized. In the code the per-cell penalties are summed and added to E ; the weight is $\lambda = \text{lambda_penalty}$.

Gradient. The gradient of the weighted variance (15) is

$$\nabla_u \text{Var}(u) = \frac{2}{W} v \odot (u - \mu \mathbf{1}), \quad (18)$$

where the terms from the μ -dependence cancel: $\partial\mu/\partial u_k = v_k/W$ and $\sum_j v_j (u_j - \mu) = 0$ on any u . For the *fixed* target T is constant, so

$$\nabla_u P = -\frac{\lambda}{T_{\text{eff}}} \nabla_u \text{Var}(u). \quad (19)$$

For the *adaptive* target $T = \mu(1 - \mu)$ depends on u through $\partial T/\partial u = (1 - 2\mu)v/W$, giving the quotient-rule form

$$\nabla_u P = -\lambda \left[\frac{\nabla_u \text{Var}(u)}{T_{\text{eff}}} - \frac{\text{Var}(u)}{T_{\text{eff}}^2} (1 - 2\mu) \frac{v}{W} \right]. \quad (20)$$

Code: crispness penalty and gradient

```
src/optimization/pgd_optimizer.py: compute_energy / compute_gradient (penalty loop)
mu = (v @ u) / W; var_w = ((center * v) @ center) / W (center = u - mu)
grad_var = (2/W) * (v * center)
fixed: G += -lambda * (grad_var / T_eff)
adaptive: G += -lambda * (grad_var/T_eff - (var_w/T_eff**2)*(1-2*mu)*(v/W))
```

Remark 6.1 (Operating regime after the energy fix). The $\sim 25\times$ scale shift from the double-well correction (Remark 4.2) rebalances λ against the well and Dirichlet terms. Empirically, on the corrected energy λ becomes an effective lever where it was inert before: a moderate $\text{lambda_penalty} \approx 5$ resolves the high- N “runt” cell that the buggy energy manufactured, and seeded initialization becomes mandatory (random initialization now stalls at the symmetric state). Both effects are measured in `docs/experiments/02-corrected-energy-highn-validation/`; they are consequences of the energy *scale*, not of the formulas derived here.

7 Assembled energy and gradient

Collecting Sections 3–6, with $q_k = u_k \odot (1 - u_k)$, the discretized Phase 1 energy and its gradient (column k acting on cell k) are

$$E(u) = \sum_{k=1}^N \left[\varepsilon u_k^\top K u_k + \frac{1}{\varepsilon} q_k^\top M q_k \right] + \lambda \sum_{k=1}^N \left(1 - \frac{\text{Var}(u_k)}{T_{\text{eff}}} \right), \quad (21)$$

$$\nabla_{u_k} E = \underbrace{2\varepsilon K u_k}_{\text{Dirichlet}} + \underbrace{\frac{2}{\varepsilon} (M q_k) \odot (1 - 2u_k)}_{\text{double well}} + \underbrace{\nabla_{u_k} P(u_k)}_{(19) \text{ or } (20)}. \quad (22)$$

The gradient (22) is the unconstrained (Euclidean) gradient the optimizer then projects onto the tangent of the sum-to-one and equal-area constraints (`src/optimization/projection.py`); the constraint handling is outside the scope of this energy derivation.

Code: assembled energy and gradient

```
src/optimization/pgd_optimizer.py:  compute_energy(x, return_components) returns
{total, grad, interface, penalty}; compute_gradient(x) returns the stacked  $\nabla E$  reshaped
to  $\mathbb{R}^{|V| \times N}$ .
```

Summary table

Quantity	Discretized form	Method	Implementing function
Dirichlet term (7)	$\varepsilon u_k^\top K u_k$	analytical	<code>compute_energy</code>
its gradient (8)	$2\varepsilon K u_k$	analytical	<code>compute_gradient</code>
Double-well term (10)	$\varepsilon^{-1} q_k^\top M q_k, \quad q_k = u_k(1 - u_k)$	analytical	<code>compute_energy</code>
its gradient (12)	$\frac{2}{\varepsilon} (M q_k) \odot (1 - 2u_k)$	analytical	<code>compute_gradient</code>
Weighted variance (15)	$\frac{1}{W} \sum_j v_j (u_j - \mu)^2$	analytical	<code>compute_energy</code>
its gradient (18)	$\frac{2}{W} v \odot (u - \mu \mathbf{1})$	analytical	<code>compute_gradient</code>
Crispness penalty (17)	$\lambda(1 - \text{Var}/T_{\text{eff}})$	analytical	<code>compute_energy</code>
its gradient (fixed / adaptive)	(19) / (20)	analytical	<code>compute_gradient</code>
Γ -limit constant (14)	$c_W = 2 \int_0^1 \sqrt{W} = \frac{1}{3}$	analytical	— (theory)

All gradients are analytical and consistent with the corresponding energy to machine precision (finite-difference relative error $\approx 10^{-8}$). The formulas above transcribe the corrected code; the superseded (buggy) double-well discretization is recorded in Remark 4.2 and audited in `docs/reference/phase1_energy_discretization_bug.md`.

References

- [1] Benjamin Bogosel and Edouard Oudet. Partitions of minimal length on manifolds. *Experimental Mathematics*, 2023. Reference for the Phase 1 (Γ -convergence relaxation) and Phase 2 (direct perimeter minimization) methodology implemented in `surface-partition`.
- [2] Luciano Modica. The gradient theory of phase transitions and the minimal interface criterion. *Archive for Rational Mechanics and Analysis*, 98(2):123–142, 1987. Classical Γ -convergence result for the Modica–Mortola functional; source of the interface constant $c_W = 2 \int_0^1 \sqrt{W}$ used in the Phase 1 energy.